INCH-POUND

MIL-DTL-81581/4B
24 July 2006
SUPERSEDING
MIL-H-81581/4A
10 December 1982

DETAIL SPECIFICATION SHEET

HOSE ASSEMBLIES, BREATHING OXYGEN, LOW PRESSURE, WITH/WITHOUT CONNECTIONS FOR AN INTEGRATED COMMUNICATION SYSTEM

Reactivated after 24 July 2006 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the hose assemblies described herein shall consist of this specification sheet and MIL-DTL-81581.

1. SCOPE

1.1 Scope. This specification sheet covers the detail requirements for two types of low pressure oxygen hose assemblies which extend from the oxygen mask to a connector or chest-mounted regulator.

1.2 Classification. Hose and hose assemblies covered by this specification sheet are of the following types, as specified (see 6.2):

Type I - Hose assembly without integrated communication cable (MS90339-2)

Type II - Hose assembly with integrated communication cable

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 491000B120-3, Highway 547, Lakehurst, NJ 08733-5100 or emailed to thomas.omara@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specification forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of this document are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-81581	-	Hose Assemblies, Breathing Oxygen and Air: General Specification for
MS90339	-	Hose Assembly, Oxygen Mask to Connector. (Inactive)

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following drawing forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of this document are those cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

280AS100	-	Hose Assembly Index, Breathing Oxygen, Low Pressure-With/Without Connections for an Integrated Communication System.

(Copies of the above drawing are available from the Naval Air Technical Data and Engineering Service Command (NATEC), P.O. Box 357031, NASNI, Bldg. 90, San Diego, CA 92135-7031 or at https://webjedmics.natec.navy.mil.)

2.2 Order or precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-DTL-81581 and as specified herein.

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3.2 Qualification. The hoses furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.1.1). Qualification shall be in accordance with groups as listed on NAVAIR Drawing 280AS100. Qualification approval of any individual hose listed within a group shall automatically qualify a manufacturer for all other items listed within the group, provided the conditions set forth in MIL-DTL-81581, 6.3, Qualification, are complied with.

3.3 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.2.

3.4 Design and construction. The hose assembly shall be in accordance with the general specification (except that the bore need not be smooth) and in accordance with NAVAIR Drawing 280AS100.

3.5 Performance. Unless otherwise specified herein, the performance of the hose assembly shall be as specified in the general specification, consistent with the applicable performance requirements for the applicable tests and inspections specified herein.

3.5.1 Polarity (color code) and electrical continuity (Type II only). When the hose assembly is tested as specified in the general specification, the resistance between all points designed to be electrically connected, shall be as specified in the general specification, with an allowable resistance of 0.03 ohms for each lineal foot of electrical wire in the hose.

3.5.2 Elongation. When tested as specified in the general specification, the increase in length of the hose shall be not greater than 50 percent for Type I and 15 percent for Type II. The permanent set shall be not greater than 5 percent of the original hose length when determined 2 minutes after removal of the load.

3.5.3 Strength.

3.5.3.1 Tensile load. When tested as specified in the general specification, there shall be no indication of damage to any portion of the hose assembly as a result of this test.

3.5.3.2 Static load. When tested as specified in the general specification, the hose shall not show a deflection of greater than 25 percent of original value on the outside diameter. With the load removed, the outside diameter of the hose shall be within 5 percent of original value.

3.5.4 Restraint cord elongation and tensile strength. When the applicable control drawing specifies a restraint cord, the restraint cord shall have a 30 percent maximum elongation and a minimum tensile strength of 70 pounds when tested as specified in the general specification.

3.5.5 Weight. When tested as specified in the general specification, the weight of the hose assembly shall be not greater than the following:

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Type	Maximum weight in ounces
I	5.0
II	9.0

4. VERIFICATION

4.1 Sampling and inspections. Sampling and inspections shall be in accordance with the general specification and as specified herein.

4.1.1 Qualification inspection. The qualification inspection of the hose assembly shall consist of the following examinations and tests from the general specification:

Visual examination
Polarity (color code) and electrical continuity, Procedure I (Type II only)
Insulation resistance, Procedure I (Type II only)
Elongation, Procedure II
Delamination, Procedure I
Odor, Procedure I
Cleanliness, Procedure I and II
Leakage, Procedure II
Flexibility, Procedure IV (for environmental conditions, Procedure II)
Minimum burst pressure, Procedure II
Tensile load, Procedure V
Restraint cord elongation and tensile strength
Static load, Procedure I
Low temperature, Procedure I
High temperature, Procedure I
Flexibility endurance, Procedure II
Ozone resistance, Procedure II
Abrasion, Procedure I
Weight, Procedure I

4.1.2 First article inspection. The first article inspection of the hose shall consist of the applicable examinations and tests specified in the general specification with the applicable procedure as specified herein.

4.1.2.1 First article samples. The first article sample shall consist of four hose assemblies of the part number contracted for.

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4.1.3 Conformance inspection. Conformance inspection shall consist of the following tests from the general specification. The sample size and acceptance criteria shall be as specified in the general specification.

Visual examination
Polarity (color code) and electrical continuity , Procedure I (Type II only)
Insulation resistance, Procedure I (Type II only)
Elongation, Procedure II
Delamination, Procedure I
Odor, Procedure I
Cleanliness, Procedure I and II
Leakage, Procedure II
Flexibility, Procedure IV

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the general specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hose assemblies covered by this detail specification are intended for use with the oxygen mask and a connector or chest-mounted regulator in supplying breathing oxygen to aircrewmen in the operation of aircraft.

6.2 Acquisition requirements. Acquisition requirements should be in accordance with the general specification.

6.3 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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